Jatin Chowdhury

Audio Signal Processing Engineer

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Education

Massachusetts Institute of Technology — *Cambridge, MA*
Ph.D in Electrical Engineering and Computer Science — *Sept. 2025 - PRESENT*
- Relevant research: real-time circuit simulation, audio software development, signal processing.
- Relevant coursework: numerical algorithms, TinyML.

Stanford University, Center for Computer Research in Music and Acoustics — *Palo Alto, CA*
M.A. in Music, Science, and Technology — *Sept. 2018 - June 2020*
- Relevant coursework: advanced digital signal processing, machine learning, spatial audio, music recording.
- Research: nonlinear signal processing, virtual analog modelling, modal signal processing.

University of Southern California — *Los Angeles, CA*
B.S. in Electrical Engineering, Physics (Minor), Music Recording (Minor) — *Aug. 2014 - May 2018*
- Relevant coursework: signal processing, circuit design, digital logic, electromagnetics, software design.

Experience

Chowdhury DSP — *Renton, WA*
Audio Signal Processing Engineer — *Oct. 2021 - PRESENT*
- Developed signal processing algorithms for real-time audio processing and synthesis.
- Contributed to open-source audio plugins and libraries.

Tesla Motors — *Palo Alto, CA*
Audio Test Engineer — *Aug. 2020 - Oct. 2021*
- Developed audio system signal flow layouts for vehicles using AudioWeaver.
- Developed end-of-line tests to ensure vehicle audio system quality using Python and C++.
- Contributed to the signal processing and tuning for the vehicle pedestrian warning speaker.

Publications & Presentations

Performance-Oriented Wave Digital Circuit Emulation — *Sept. 2026*
Proc. of the 29th International Conference on Digital Audio Effects — *Cambridge, USA*
- Co-authored with Mark Rau.
- Available on the DAFx Archives.

Performance Engineering for Modern DSP — *Sept. 2025*
Signal Processing Summit — *Milpitas, CA*
- Discussed performance engineering based on modern advancements in hardware technology.
- Discussed ideas for teaching novice engineers about performance.

Inference-Time Structured Pruning for Real-Time Neural Network Audio Effects — *Sept. 2025*
Proc. of the 28th International Conference on Digital Audio Effects — *Ancona, IT*
- Co-authored with Christopher Johann Clarke.
- Available on the DAFx Archives.

RTNeural: Fast Neural Inferencing for Real-Time Systems — *June 2021*
ArXiv E-Prints: Audio and Speech Processing — *Online*
- Presents a neural network inferencing library for real-time systems.
- Available on the ArXiv.

Real-Time Physical Modelling for Analog Tape Machines — *Sept. 2019*
Proc. of the 22nd International Conference on Digital Audio Effects — *Birmingham, UK*
- Oral presentation at the DAFx-2019 conference.
- Available on the DAFx Archives.

Skills

Programming Languages C/C++, Python, Jai, Bash, MATLAB, Faust, LaTeX
Tools/Frameworks CMake, Git, Linux CLI, Visual Studio, Xcode, CI/CD, JUCE, CLAP